on our earlier discussion, because there, we were concerned only with horizontal motion, motion that was unaffected by the vertical gravity experienced by all participants. But as an important point of principle, the link Einstein found between gravity and acceleration means, once again, that we are justified only in considering stationary those observers who feel *no* forces whatsoever.

Having forged the link between gravity and acceleration, Einstein was now ready to take up Newton's challenge and seek an explanation of how gravity exerts its influence.

Warps, Curves, and Gravity

Through special relativity, Einstein showed that every observer cuts up spacetime into parallel slices that he or she considers to be all of space at successive instants of time, with the unexpected twist that observers moving relative to one another at constant velocity will cut through spacetime at different angles. If one such observer should start accelerating, you might guess that the moment-to-moment changes in his speed and/or direction of motion would result in moment-to-moment changes in the angle and orientation of his slices. Roughly speaking, this is what happens. Einstein (using geometrical insights articulated by Carl Friedrich Gauss, Georg Bernhard Riemann, and other mathematicians in the nineteenth century) developed this idea—by fits and starts—and showed that the differently angled cuts through the spacetime loaf smoothly merge into slices that are *curved* but fit together as perfectly as spoons in a silver-ware tray, as schematically illustrated in Figure 3.8. *An accelerated observer carves spatial slices that are warped.*

With this insight, Einstein was able to invoke the equivalence principle to profound effect. Since gravity and acceleration are equivalent, Einstein understood that gravity itself must be nothing but warps and curves in the fabric of spacetime. Let's see what this means.

If you roll a marble along a smooth wooden floor, it will travel in a straight line. But if you've recently had a terrible flood and the floor dried with all sorts of bumps and warps, a rolling marble will no longer travel along the same path. Instead, it will be guided this way and that by